

Sri Lanka Institute of Information Technology



IT3021 - Data warehousing and Business Intelligence
Year 3 Semester 1

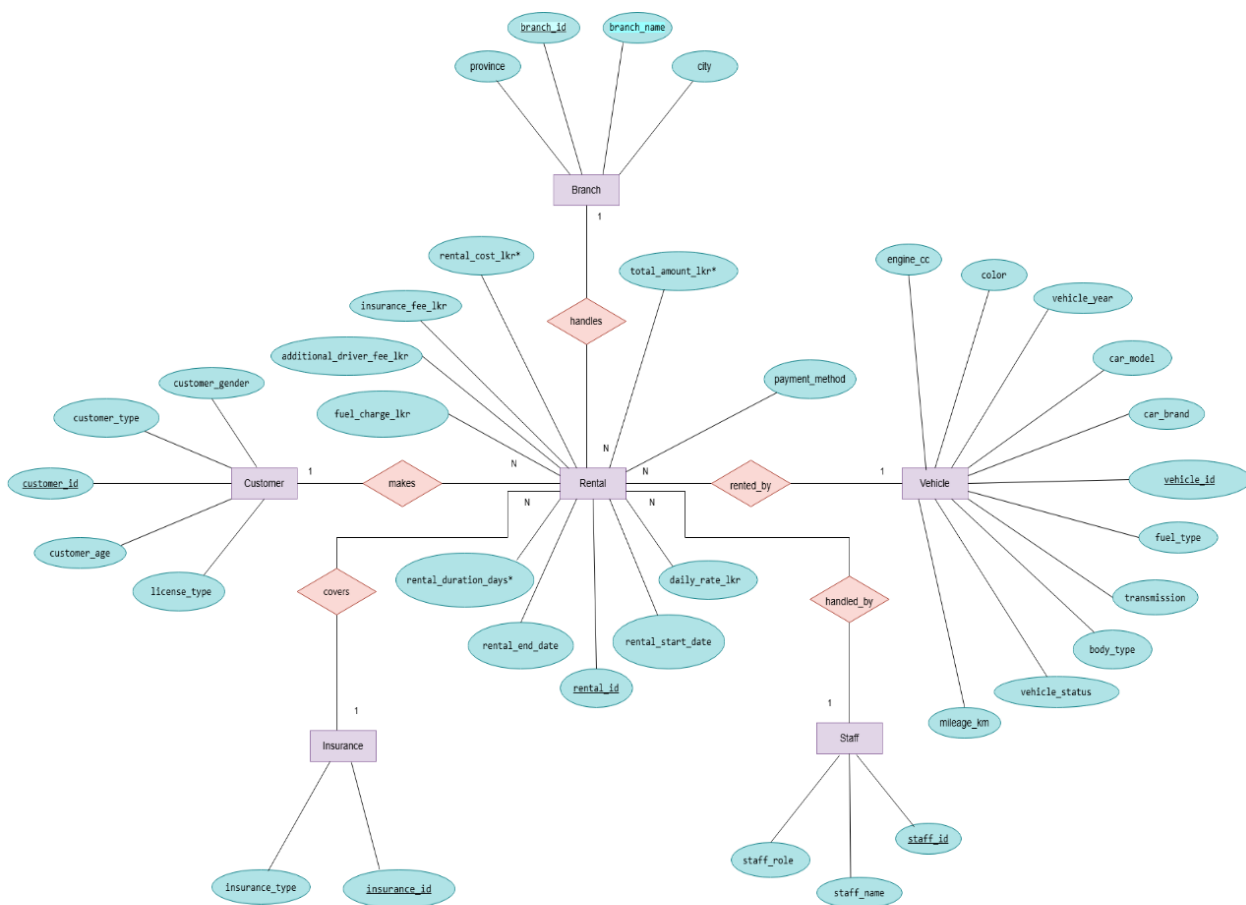
Assignment 02

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IT23714434

Step 1: Data source for the Assignment 2

The data source used for Assignment 2 is the **Car Rental Data Warehouse** originally built in Assignment 1. The data warehouse is implemented in Microsoft SQL Server under the database name CarRental_DW. It follows a **star schema** design optimized for analytical queries and OLAP cube creation.

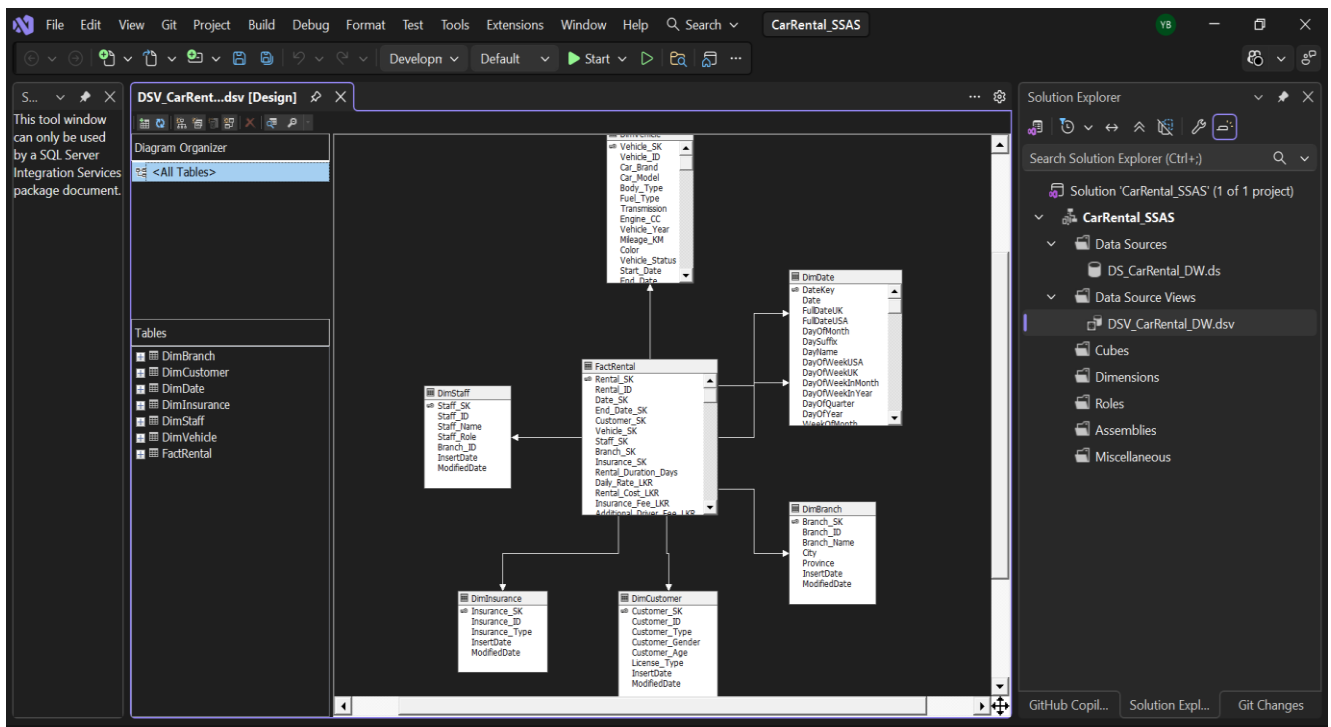
ER Diagram:

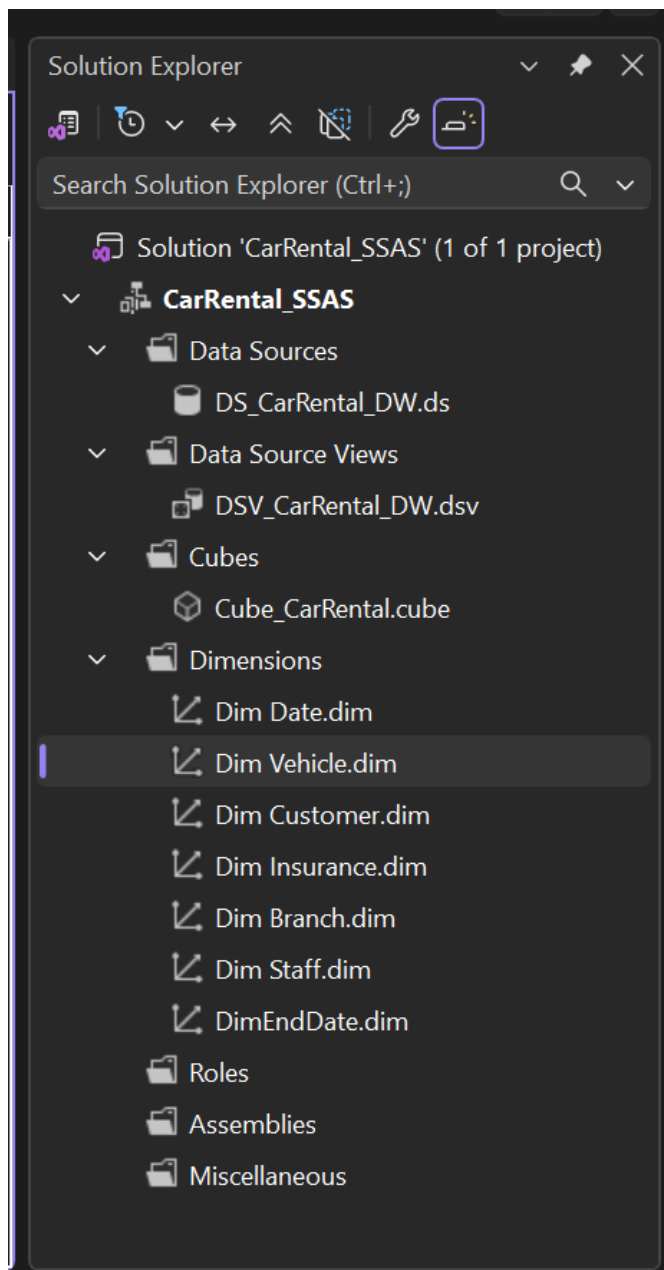


* = Derived attribute (calculated from other fields)

Step 2: SSAS Cube implementation

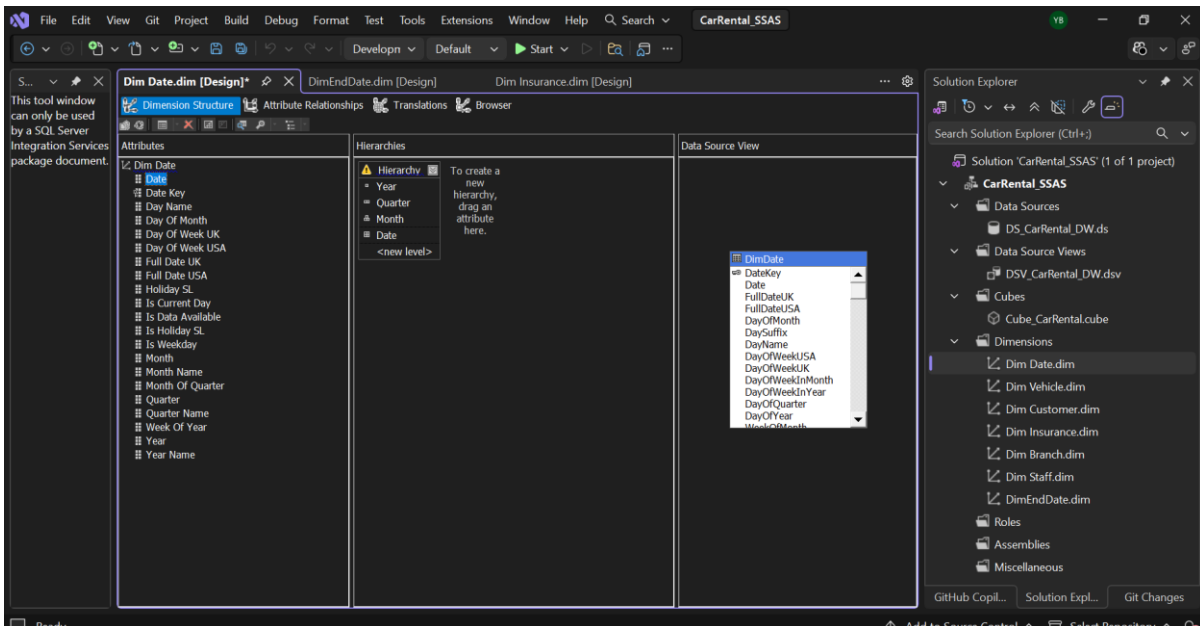
An SSAS cube named Cube_CarRental was created using SQL Server Analysis Services (Multidimensional Model) in Visual Studio with SSDT. The cube uses the CarRental_DW data warehouse as its data source. The FactRental fact table provides measures such as Total Amount LKR, Rental Duration Days, Daily Rate LKR, and Rental Count. The dimensions (DimCustomer, DimVehicle, DimBranch, DimInsurance, DimStaff, and DimDate) are connected via surrogate keys (Customer_SK, Vehicle_SK, etc.). A **Date Hierarchy** (Year → Quarter → Month → Date) was implemented in the DimDate dimension, and a **Vehicle Hierarchy** (Car Brand → Car Model → Vehicle_ID) was created in DimVehicle to enable drill-down analysis. A separate role-playing dimension DimEndDate was added to handle the End_Date_SK foreign key. The cube was successfully deployed to the DESKTOP-U5P1510\SQLDEV_EVAL Analysis Services instance and processed, allowing for multidimensional analysis of rental revenue, vehicle utilization, branch performance, and customer rental patterns.





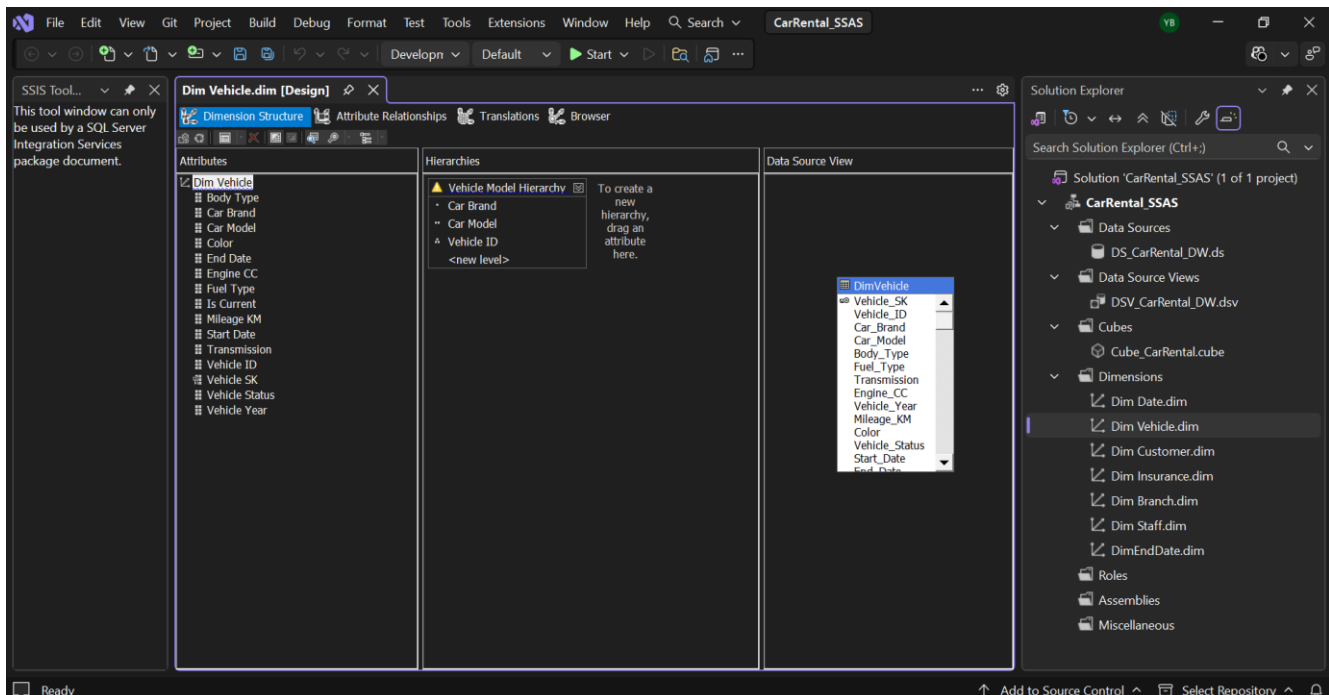
Date Hierarchy

Year → Quarter → Month → Date

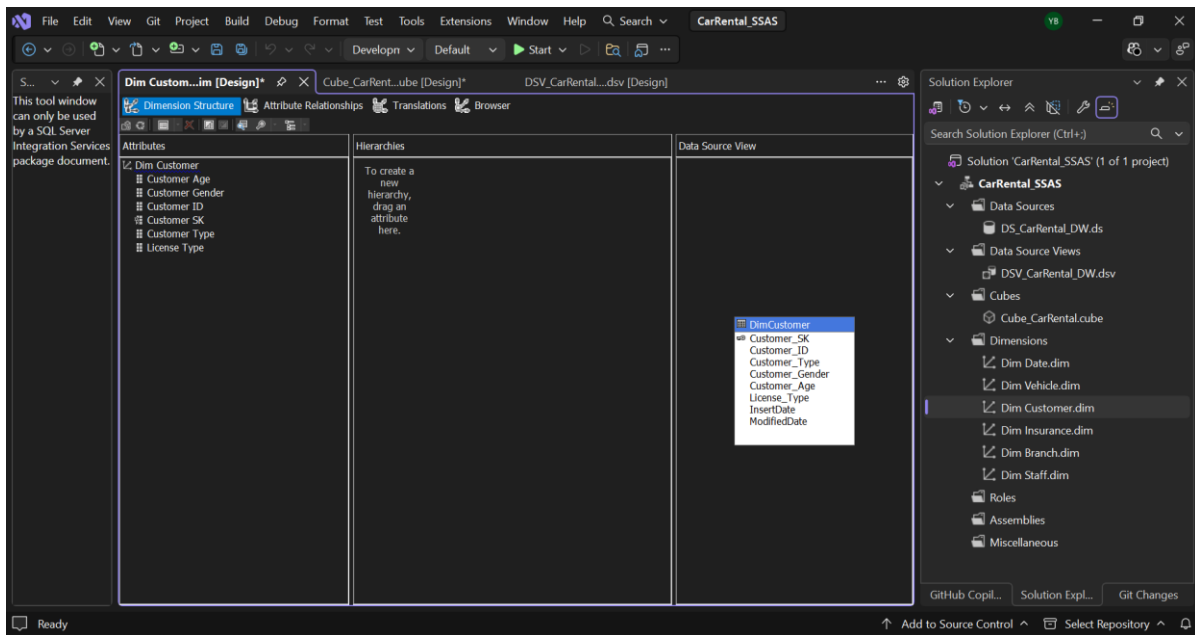


Vehicle Hierarchy

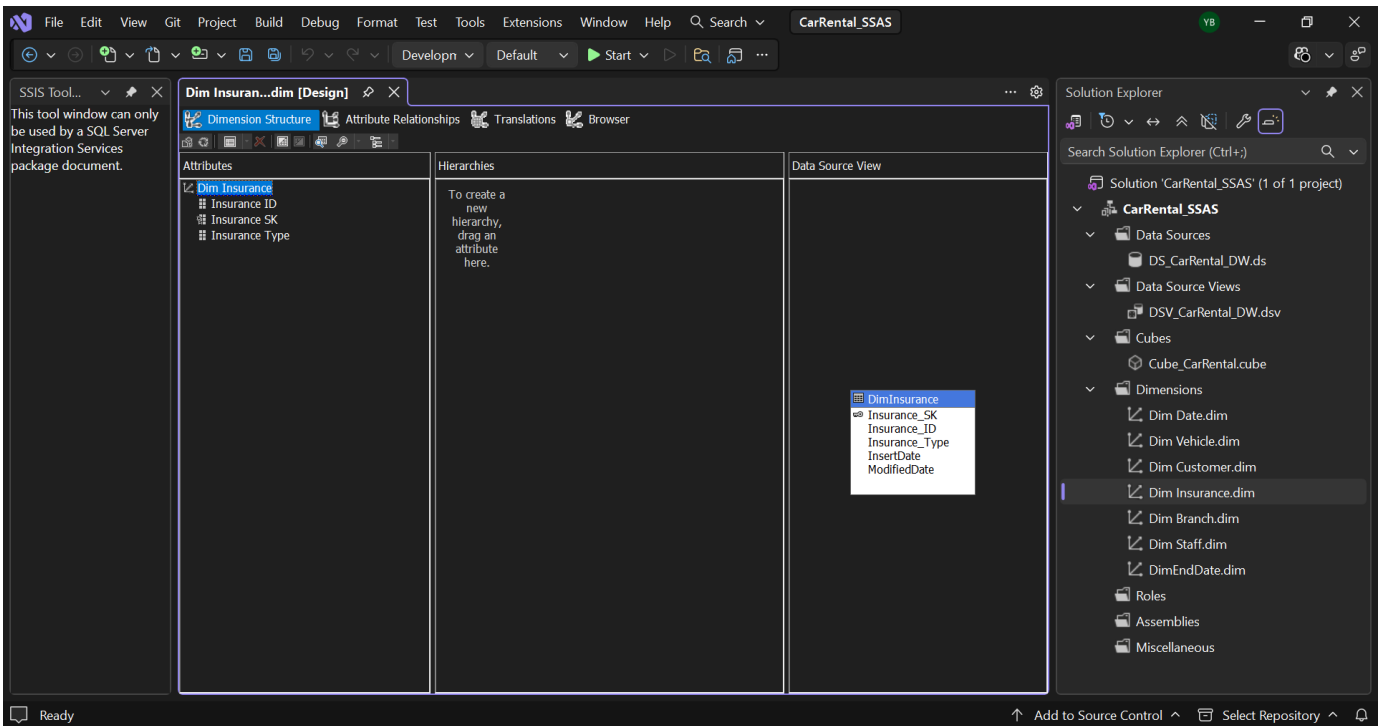
Car Brand → Car Model → Vehicle_ID



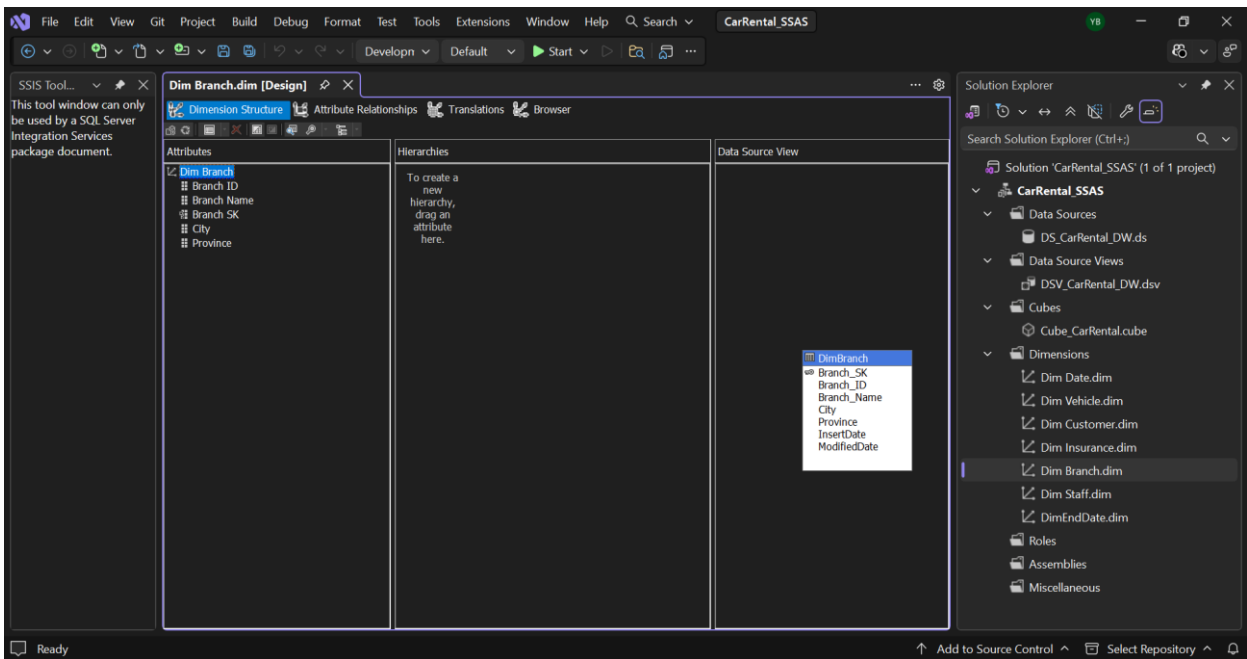
Dim Customer



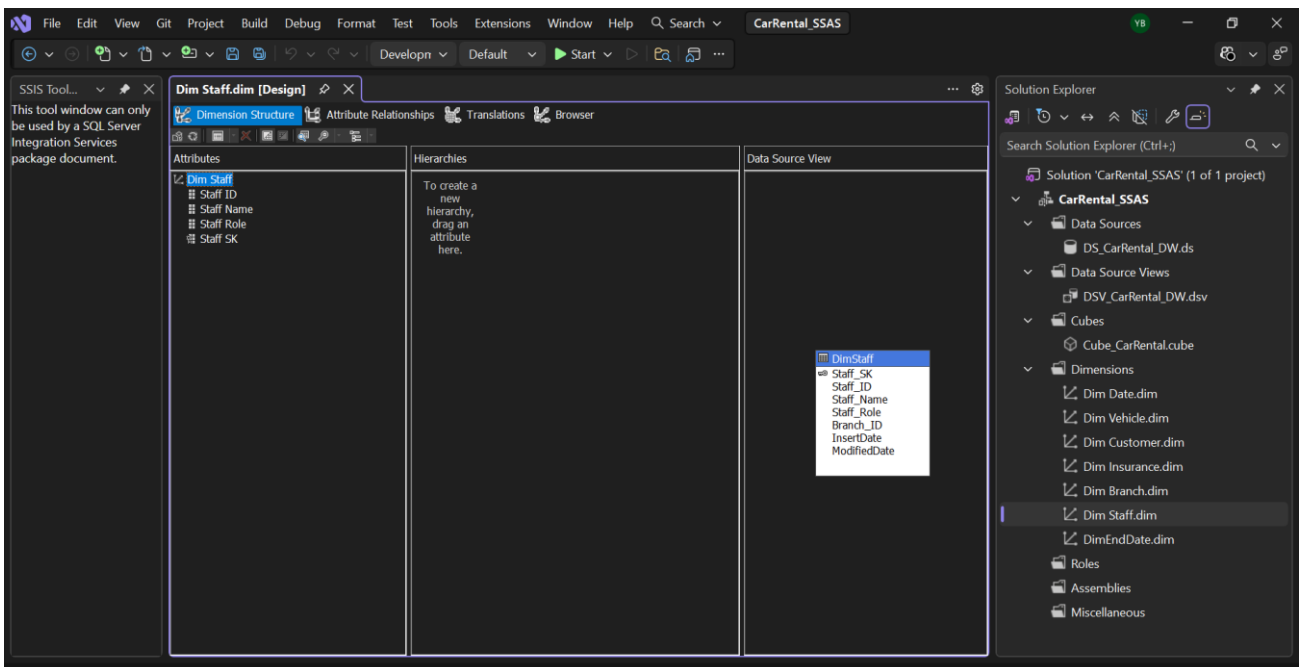
Dim Insurance



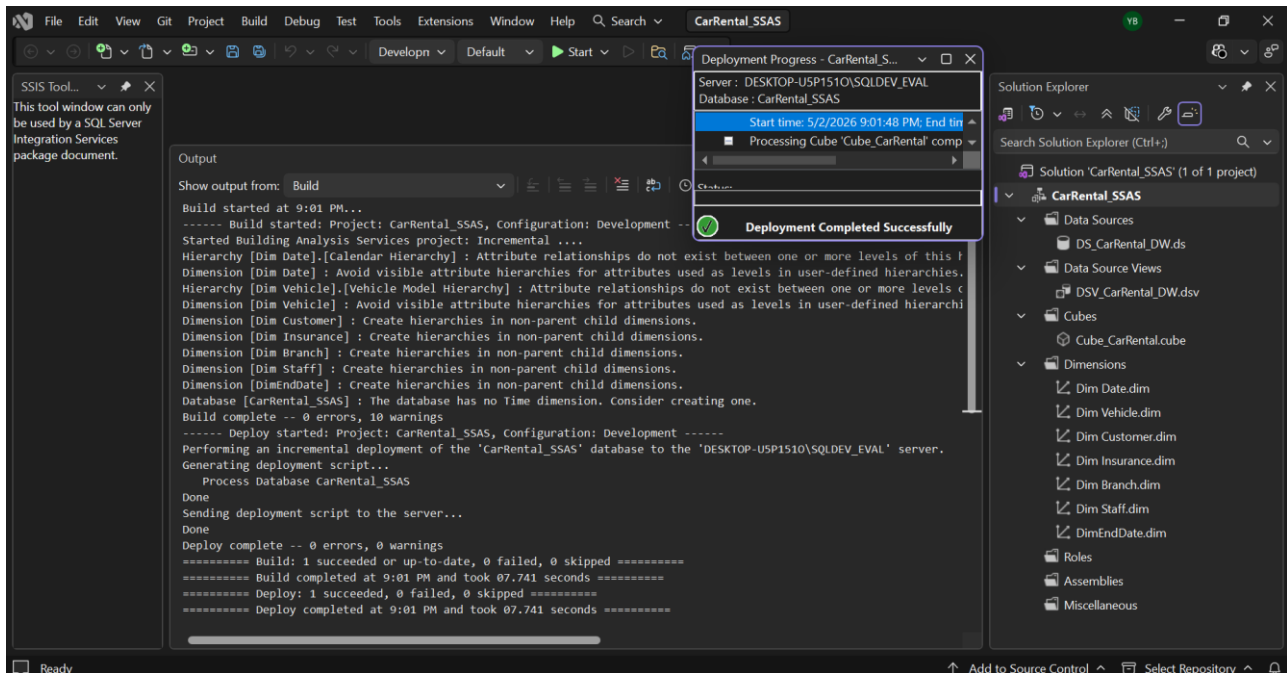
Dim Branch



Dim Staff



Cube Deployment



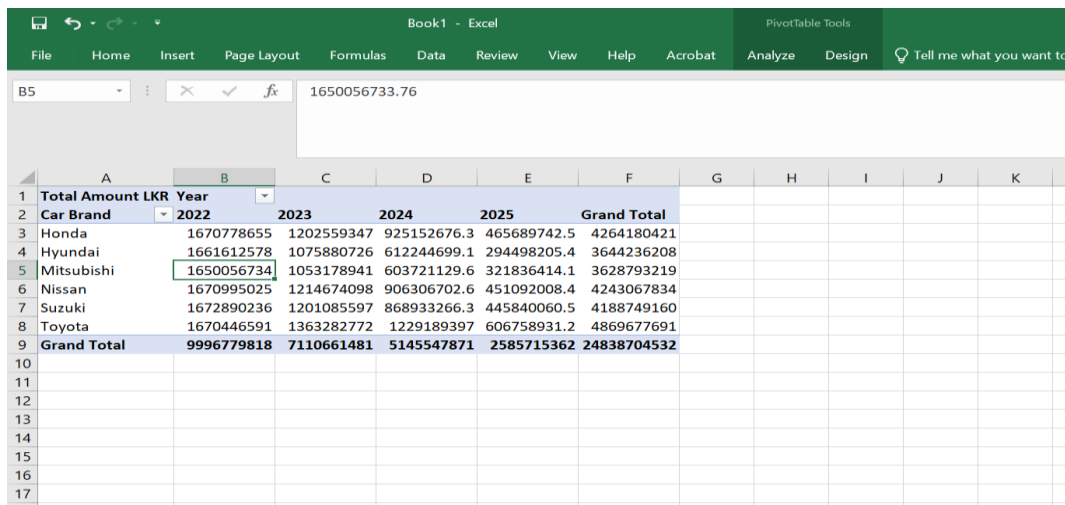
Step 3: Demonstration of OLAP operations

Connecting to the Cube

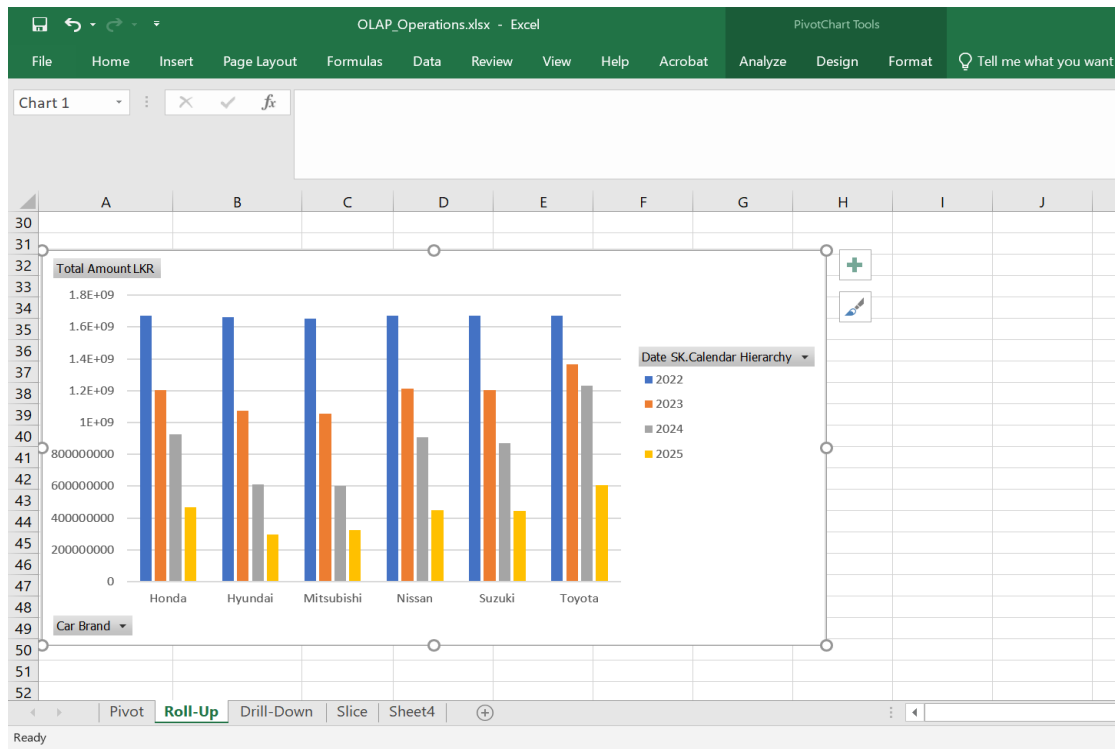
1. Opened a new Excel workbook.
2. Navigated to **Data** → **Get Data** → **From Database** → **From Analysis Services**.
3. Provided the server name: DESKTOP-U5P1510\SQLDEV_EVAL (the Analysis Services instance where the cube is deployed).
4. Selected the database CarRental_SSAS and the cube cube_CarRental.
5. Clicked **Finish** and then **OK** to create a blank PivotTable on a new worksheet (named ConnectionTest). Verified connectivity by adding Total Amount LKR to Values, Car Brand to Rows, and Year to Columns. After confirming, the test sheet was deleted.

1. Roll-up Operation (Worksheet: Roll-up)

For the roll-up operation, I built a PivotTable with 'Car Brand' on Rows, and both 'Year' and 'Quarter' on Columns (using the date hierarchy). The values were 'Total Amount LKR'. The detailed view displayed quarterly revenue per brand. Then I removed the 'Quarter' field from the Columns area, leaving only 'Year'. This aggregated the data from quarter level to year level, which is a classic roll-up operation

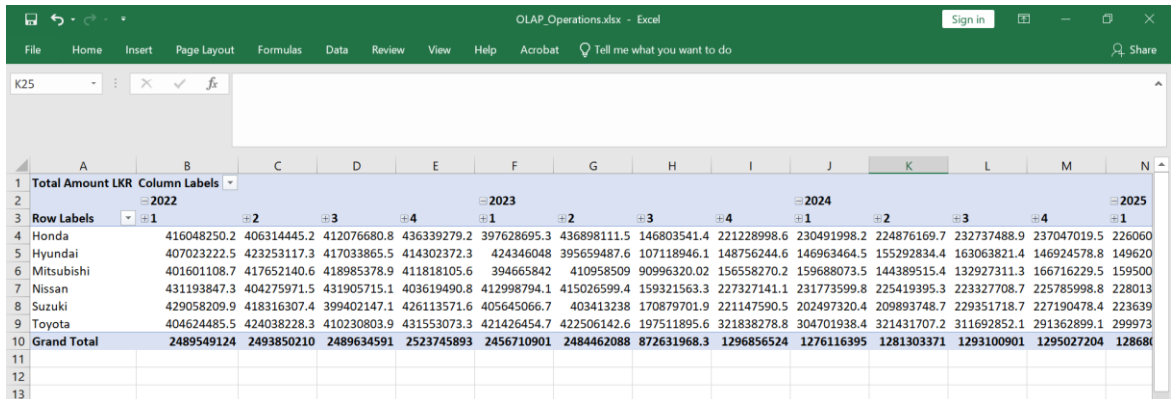


Total Amount LKR	Year				
Car Brand	2022	2023	2024	2025	Grand Total
Honda	1670778655	1202559347	925152676.3	465689742.5	4264180421
Hyundai	1661612578	1075880726	612244699.1	294498205.4	3644236208
Mitsubishi	1650056734	1053178941	603721129.6	321836414.1	3628793219
Nissan	1670995025	1214674098	906306702.6	451092008.4	4243067834
Suzuki	1672890236	1201085597	868933266.3	445840060.5	4188749160
Toyota	1670446591	1363282772	1229189397	606758931.2	4869677691
Grand Total	9996779818	7110661481	5145547871	2585715362	24838704532



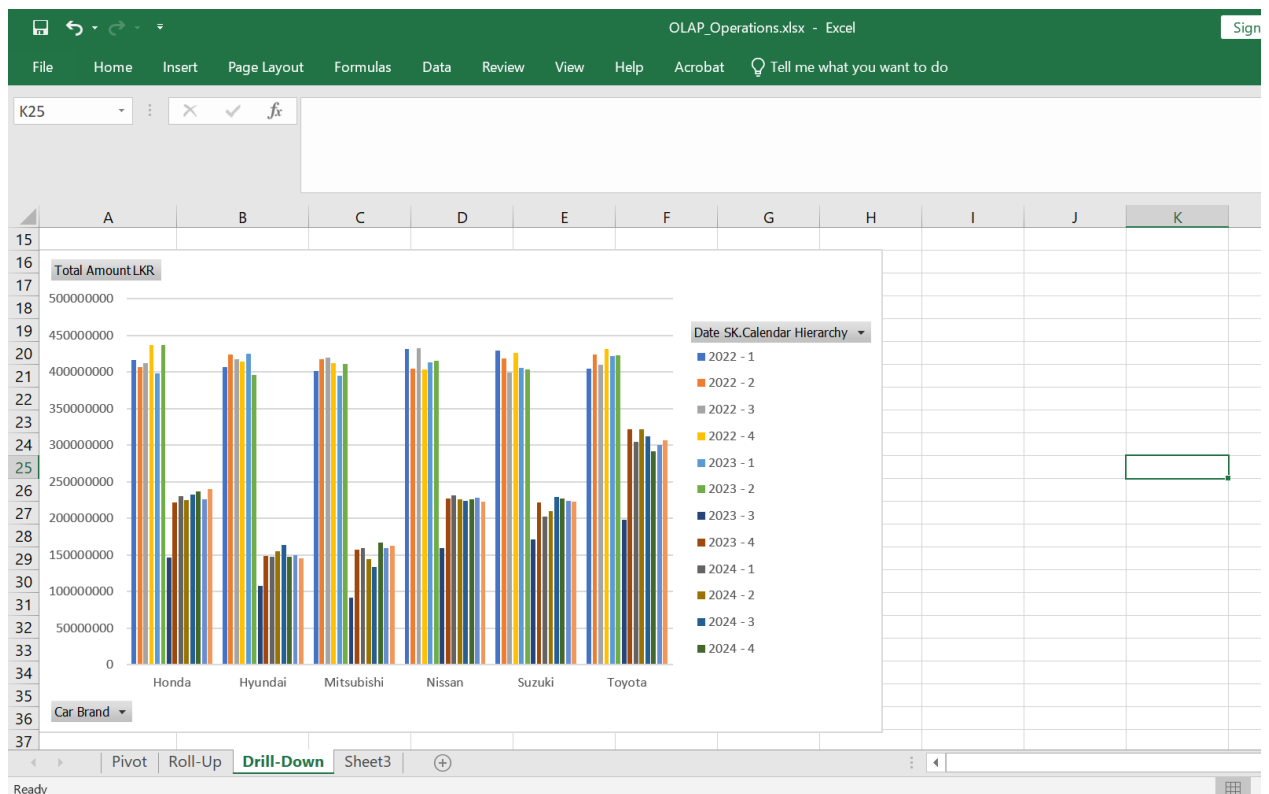
2. Drill-Down Operation (Worksheet: Drill-Down)

I demonstrated drill-down using the date hierarchy and the vehicle hierarchy. In the PivotTable, I placed 'Car Brand' and 'Car Model' on Rows (vehicle hierarchy), and the date hierarchy ('Year' → 'Quarter' → 'Month') on Columns. The measure was 'Total Amount LKR'. Starting from the year level, I clicked the expand icons ('+') to drill down first to quarters and then to months



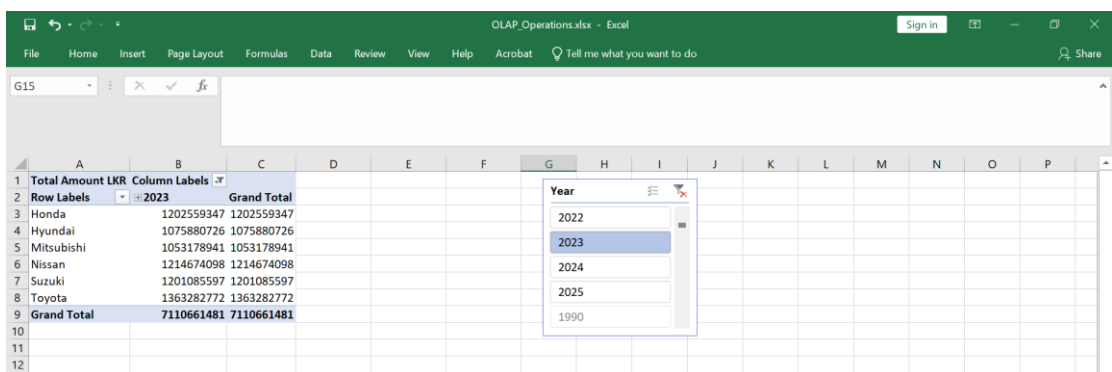
The screenshot shows an Excel PivotTable titled 'OLAP_Operations.xlsx - Excel'. The PivotTable is set to show 'Total Amount LKR' as the measure. The rows are labeled with 'Car Brand' and 'Car Model'. The columns are labeled with 'Year' and 'Quarter'. The data is summarized by year (2022, 2023, 2024, 2025) and quarter (1, 2, 3, 4). The table includes a 'Grand Total' row at the bottom.

	2022	2023	2024	2025
Row Labels	1	2	3	4
Honda	416048250.2	406314445.2	412076680.8	436339279.2
Hyundai	407023222.5	423253117.3	417033865.5	414302372.3
Mitsubishi	401601108.7	417652140.6	418985378.9	411818105.6
Nissan	431193847.3	404275971.5	431905715.1	403619490.8
Suzuki	429058209.9	418316307.4	399402147.1	426113571.6
Toyota	404624485.5	424038228.3	410230803.9	431553073.3
Grand Total	2489549124	2493850210	2489634591	2523745893



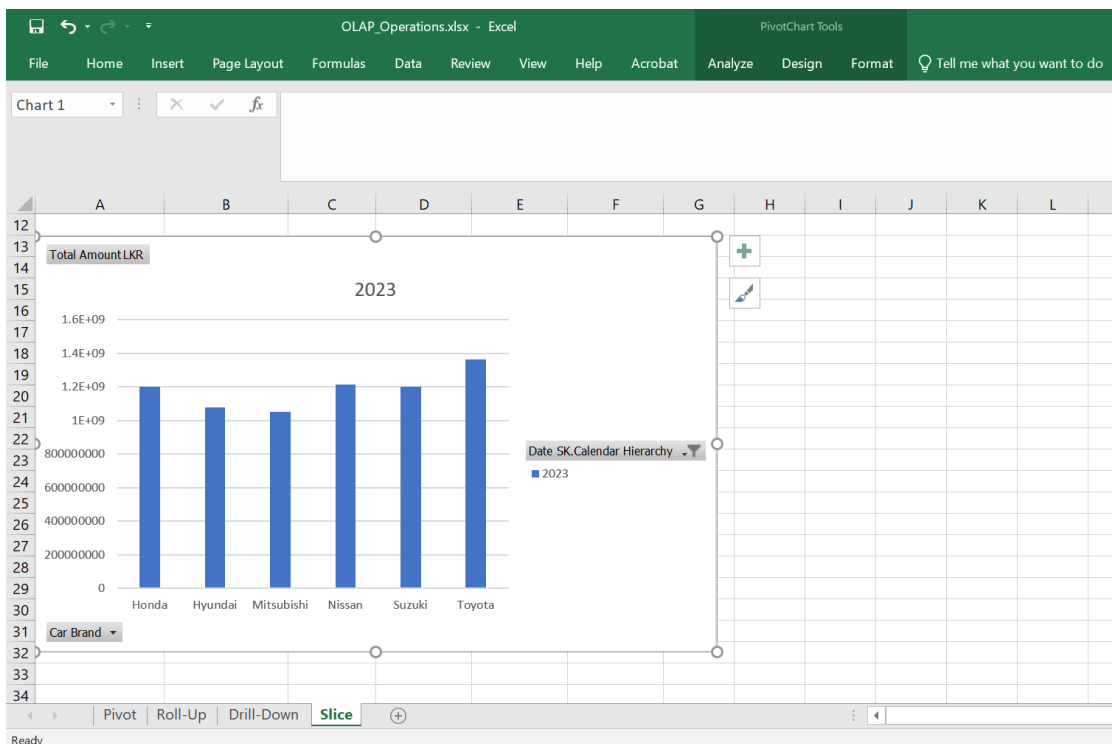
3. Slice Operation (Worksheet: Slice)

To illustrate the slice operation, I created a PivotTable with 'Car Brand' on Rows, 'Year' on Columns, and 'Total Amount LKR' in Values. I then inserted a slicer for the 'Year' dimension. By selecting only the year 2023 in the slicer, the PivotTable was filtered to show revenue data exclusively for 2023. This is a single-dimension filter, commonly known as a 'slice'. A clustered column chart was added to visualise the sliced data



The screenshot shows an Excel spreadsheet with a PivotTable and a slicer. The PivotTable is located in the range A1:C9. The slicer for the 'Year' dimension is located in the range G15:I25. The PivotTable shows the following data:

Row Labels	2023	Grand Total
Honda	1202559347	1202559347
Hyundai	1075880726	1075880726
Mitsubishi	1053178941	1053178941
Nissan	1214674098	1214674098
Suzuki	1201085597	1201085597
Toyota	1363282772	1363282772
Grand Total	7110661481	7110661481

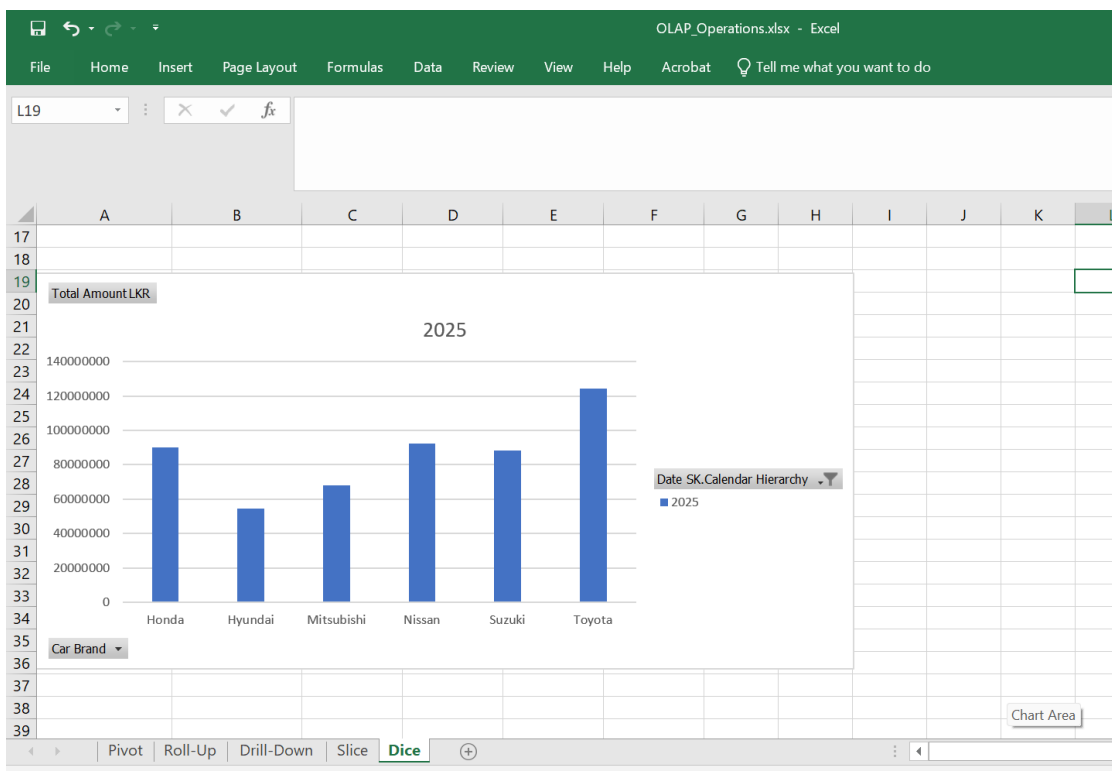


4. Dice Operation (Worksheet: Dice)

For the dice operation, I started with a PivotTable that had 'Car Brand' on Rows, 'Year' on Columns, and 'Total Amount LKR' in Values. I inserted two slicers: one for 'Year' and one for 'City' (from the 'DimBranch' dimension). Then I selected '2025' in the Year slicer and 'Colombo' in the City slicer. As a result, the PivotTable showed only the rental revenue for the year 2025 and for the city of Colombo. This is a dice operation because it applies filters on two different dimensions simultaneously

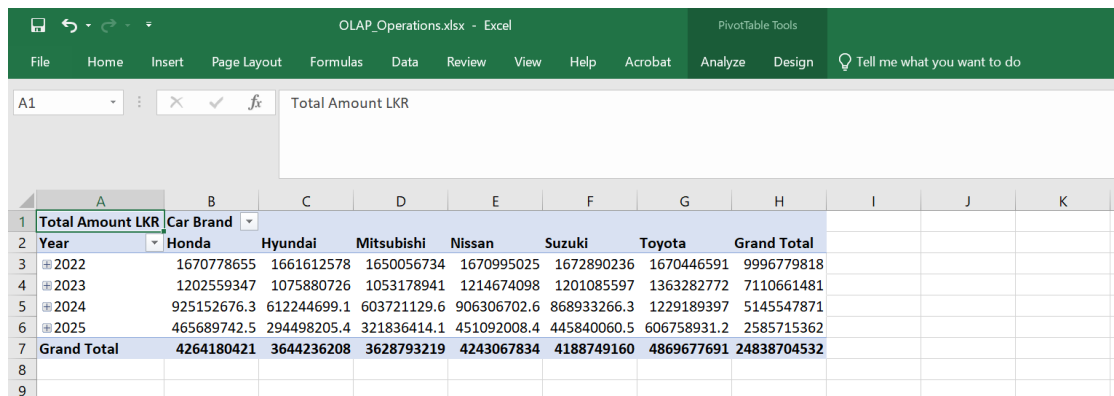
The screenshot shows an Excel PivotTable with 'Total Amount LKR' as the value field, 'Car Brand' as the row labels, and 'Year' as the column labels. The PivotTable is filtered for the year 2025 and the city of Colombo. The data is as follows:

Car Brand	Total Amount LKR
Honda	90199842.1
Hyundai	54656739.78
Mitsubishi	67973359.66
Nissan	92354579.56
Suzuki	88130455.32
Toyota	124191807.7
Grand Total	517506784.1



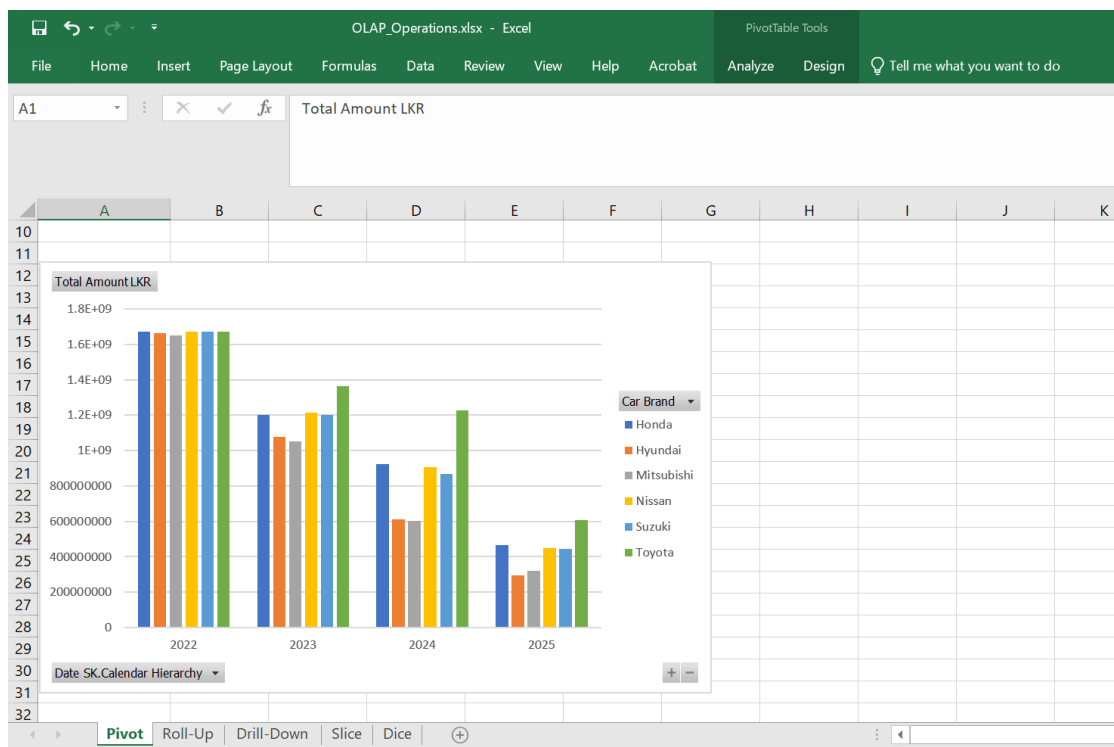
5. Pivot Operation (Worksheet: Pivot)

I created a PivotTable connected to the 'Cube_CarRental' SSAS cube. Initially, I placed 'Car Brand' on Rows, 'Year' on Columns, and 'Total Amount LKR' in Values. To demonstrate the pivot operation, I swapped the fields: I moved 'Year' to Rows and 'Car Brand' to Columns. This changed the orientation of the analysis, showing yearly totals across different car brands. A clustered column chart was added to visually compare the results before and after the swap



The screenshot shows an Excel PivotTable titled 'Total Amount LKR'. The PivotTable is structured with 'Year' on the Rows and 'Car Brand' on the Columns. The data is as follows:

Year	Honda	Hyundai	Mitsubishi	Nissan	Suzuki	Toyota	Grand Total
2022	1670778655	1661612578	1650056734	1670995025	1672890236	1670446591	9996779818
2023	1202559347	1075880726	1053178941	1214674098	1201085597	1363282772	7110661481
2024	925152676.3	612244699.1	603721129.6	906306702.6	868933266.3	1229189397	5145547871
2025	465689742.5	294498205.4	321836414.1	451092008.4	445840060.5	606758931.2	2585715362
Grand Total	4264180421	3644236208	3628793219	4243067834	4188749160	4869677691	24838704532



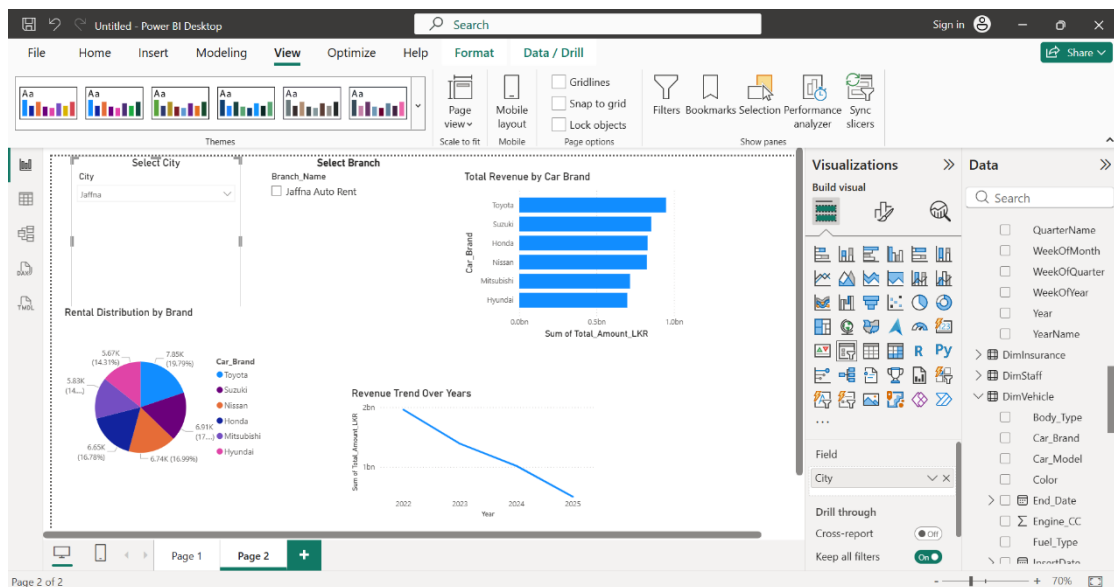
Step 4: PowerBI Reports

Report 1

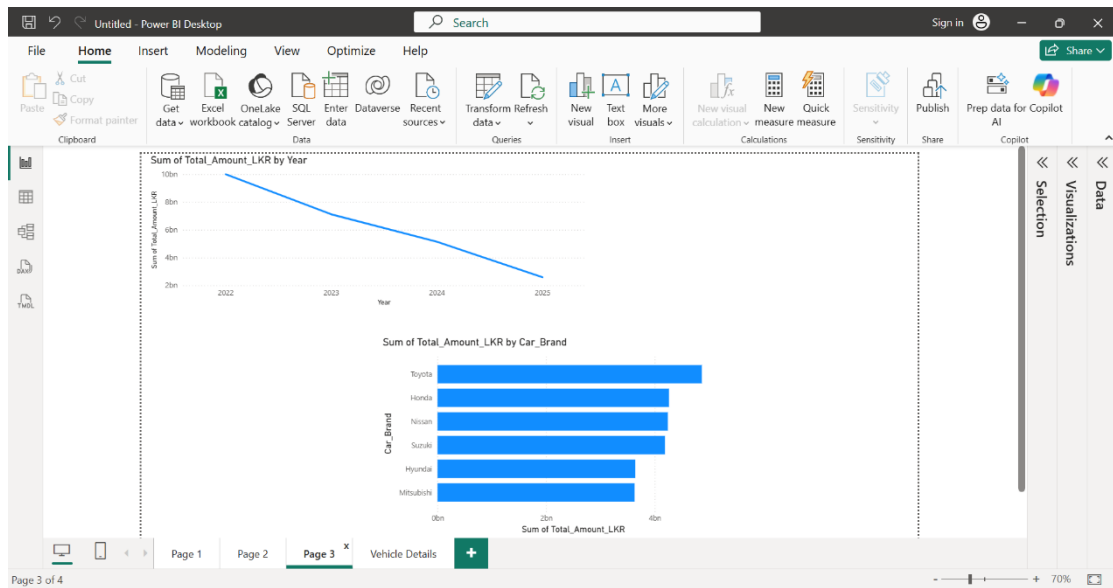
The screenshot shows the Power BI Desktop interface with a table titled "Car Rental Summary by Vehicle and Year". The table displays rental data for various car brands across the years 2023, 2024, and 2025, including total amounts and rental counts.

Year	Car_Brand	2023		2024		2025		Total	
		Total_Amount_LKR	Count of Rental_ID	Sum of Total_Amount_LKR	Count of Rental_ID	Sum of Total_Amount_LKR	Count of Rental_ID	Sum of Total_Amount_LKR	Count of Rental_ID
	Honda	1,202,559,346.80	9560	925,152,676.26	7518	465,689,742.50	3688	4,264,180,420.94	34230
	Hyundai	1,075,880,726.18	8532	612,244,699.06	4900	294,498,205.42	2398	3,644,236,208.26	29514
	Mitsubishi	1,053,178,941.28	8532	603,721,129.64	4940	321,836,414.06	2562	3,628,793,218.74	29164
	Nissan	1,214,674,097.90	9614	906,306,702.60	7302	451,092,008.44	3666	4,243,067,833.66	33930
	Suzuki	1,201,085,597.02	9768	868,933,266.28	7118	445,840,060.50	3618	4,188,749,159.84	33866
	Toyota	1,363,282,771.76	11034	1,229,189,396.74	9958	606,758,931.24	4936	4,869,677,690.64	39296
	Total	7,110,661,480.94	57140	5,145,547,870.58	41736	2,585,715,362.16	20868	24,838,704,532.08	200000

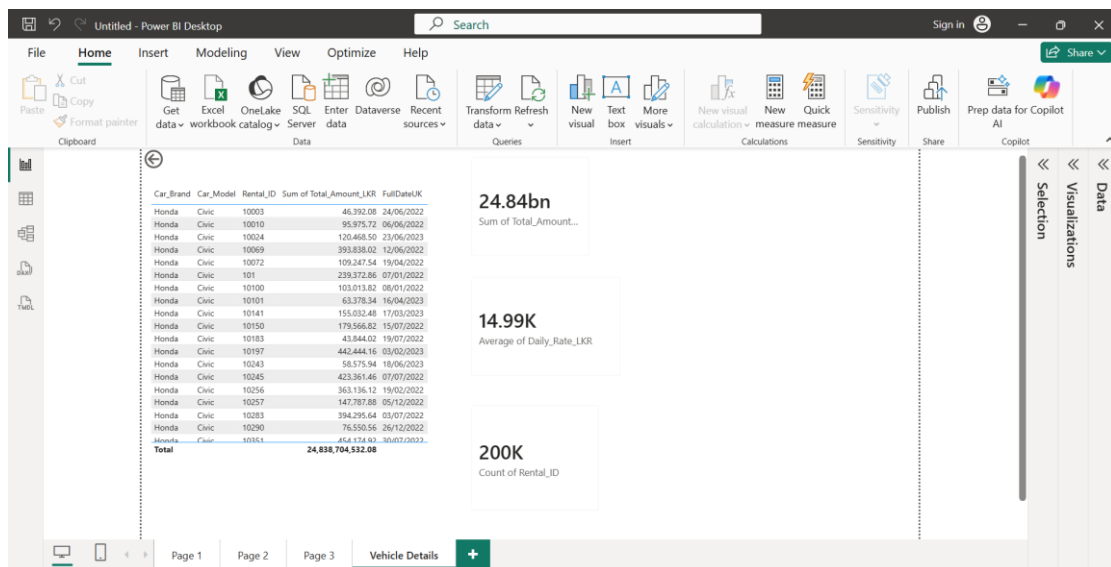
Report 2



Report 3



Report 4



Report 1 – Matrix Visual

I created a report with a Matrix visual. I connected Power BI to the 'CarRental_DW' SQL Server database. On the Matrix, I placed 'DimVehicle[Car Brand]' and 'DimVehicle[Car Model]' on Rows, and 'DimDate[Year]' and 'DimDate[Quarter]' on Columns. The values were 'Sum of Total Amount LKR' and 'Count of Rental_ID'. I formatted the matrix to show subtotals and enabled row/column groupings.

Report 2 – Cascading Slicers + Multiple Charts

I designed a report with two slicers: 'City' and 'Branch Name' (both from DimBranch). The 'Branch Name' slicer cascades based on the selected city. I added a clustered bar chart (Total Revenue by Car Brand), a pie chart (Rental Count by Brand), and a line chart (Revenue Trend over Years). All visuals interact with the slicers

Report 3 – Drill-Down Report

I built a drill-down report using a line chart with a date hierarchy (Year → Quarter → Month). By clicking on a year, the chart drills down to quarter, and then to month. I also added a vehicle brand hierarchy (Brand → Model) to demonstrate drill-down on another dimension. The drill-up arrow returns to higher levels.

Report 4 – Drill-Through Report

I created a drill-through report. The main page contains a bar chart of total revenue by car brand. I set up a detail page ('Drill-Through Details') with a table and card visuals, and configured drill-through using the 'Car Brand' field, and also showing all rentals and metrics filtered for that brand.

